Chengyang He

che14@stevens.edu | 857-272-3493 | linkedin.com/in/chengyang-he | Google Scholar

Education

Stevens Institute of Technology — Hoboken, NJ
Ph.D. in Data Science Expected: Summer 2027 — *Sep. 2022 – Present*

- **Advisor:** Dr. Ping Wang
- **Research Interests:** LLM-driven clinical reasoning, multi-agent AI systems, explainable AI, and temporal analysis in healthcare.
- **Teaching Assistant:** CS 559: Machine Learning (Spring 2024) | CS 583: Deep Learning (Spring 2025)

Boston University — Boston, MA
M.S. in Computer Science | B.A. in Mathematics & Computer Science — *2019–2021 | 2015–2019*

Research

Task as Context Prompting for Accurate Medical Symptom Coding — *IEEE/ACM CHASE 2025*

- Proposed TACO, a task-as-context prompting strategy that frames each coding instance with symptom-specific context, to address the challenge of automatically mapping free-text clinical notes to standardized medical codes with grounding mention spans, enabling the model to adapt to any target condition.
- Introduced a fine-grained evaluation framework that jointly assesses code extraction accuracy and mention-to-code alignment, providing a more complete measure of end-to-end coding performance.

SMILE-College: Sentiment Analysis of Student Mental Health Support in Colleges — *BigData 2024*

- Constructed SMILE-College (793 records), the first sentiment analysis dataset in this domain, via human-machine collaborative annotation with a coarse-to-fine LLM prompting strategy, to tackle the lack of standardized data for evaluating college mental health services.
- Benchmarked multiple LLMs on sentiment prediction, analyzed cross-category error patterns, and applied embedding-based clustering on dissatisfied responses to surface recurring systemic limitations in college mental health services.

Temporal Symptom Timeline Extraction — *Ongoing*

- Introduced a benchmark of 3,156 vaccine adverse-event reports (VAERS) with expert-validated stage-wise annotations to study the problem of recovering an ordered symptom progression from a single clinical narrative, where temporal cues are often sparse and implicit.
- Proposed an iterative generator–verifier framework in which a generator LLM proposes a bucketed temporal sequence and a verifier agent checks structural and temporal consistency, providing targeted feedback for refinement within a fixed compute budget.

MERMAID: Multi-Agent Iterative Knowledge Grounding for Veracity Assessment — *Under Review*

- Developed a multi-agent system combining memory-enhanced retrieval with iterative knowledge grounding to reliably verify claims against external knowledge, where agents collaboratively retrieve and cross-validate evidence across multiple reasoning rounds.
- Agents progressively refine their judgment through iterative evidence reconciliation, producing grounded and trustworthy verdicts on claim veracity.

FAIR-Q: Multi-Agent Framework for Assessment Question Refinement — *Under Review*

- Addressed the fairness gap in standardized assessments, which often disadvantage students with diverse backgrounds, by building a four-agent LLM pipeline (Student Simulator, Evaluation Agent, Refiner Agent, Reviewer Agent) to iteratively refine questions for inclusivity while preserving rubric consistency.
- The simulator generates realistic learner personas via clustering; the evaluator scores questions on clarity, cognitive alignment, and fairness; the refiner rewrites based on structured feedback; and the reviewer validates improvements via LLM-as-a-judge pairwise comparison.

Publications

[1] **He, Chengyang**, Wenlong Zhang, Violet Chen, Yue Ning, and Ping Wang. "Task as Context Prompting for Accurate Medical Symptom Coding Using Large Language Models." *2025 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)* (2025).

[2] Sood, Palak, **Chengyang He**, Divyanshu Gupta, Yue Ning, and Ping Wang. "Understanding Student Sentiment on Mental Health Support in Colleges Using Large Language Models." *2024 IEEE International Conference on Big Data (BigData)* (2024).

[3] Zhang, Wenlong, **Chengyang He**, Guanqun Yang, Dipankar Bandyopadhyay, Tian Shi, and Ping Wang. "Natural Language Querying on Domain-Specific NoSQL Database with Large Language Models." *2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* (2024).

[4] Cao, Yupeng, **Chengyang He**, Yangyang Yu, Ping Wang, and K. P. Subbalakshmi. "MERMAID: Memory-Enhanced Retrieval and Reasoning with Multi-Agent Iterative Knowledge Grounding for Veracity Assessment." *arXiv:2601.22361* (under review).

Work Experience

HALO, LLC — Boston, MA
Data Scientist — *Nov. 2021 – Mar. 2022*

- Collaborated with Beth Israel Deaconess Medical Center (BIDMC) to extract structured clinical insights from 5,000+ colonoscopy reports; ingested scanned documents via OCR before feeding text into the NLP pipeline.
- Designed and deployed an end-to-end IE pipeline (OCR → spaCy NER → rule-based post-processing) that boosted entity-level F1 accuracy by **28%** and reduced clinician manual review time by **45%**.
- Built and maintained gold-standard annotation datasets; managed labeling workflow in Prodigy then migrated to Doccano, cutting annual licensing costs by $1,700 while preserving annotation consistency.
- Worked closely with clinical stakeholders to iteratively refine entity schemas and post-processing rules, translating domain-specific medical terminology into robust NLP features.

Boston University — Boston, MA
Research Assistant — *Aug. 2020 – Dec. 2021*

- **Gulf War Illness Project** (Supervisor: Dr. Bangbon Koo): Modeled symptom clusters in 144 veteran self-reports using K-means, hierarchical clustering, and Gaussian-Bayesian networks; selected optimal solutions via silhouette/elbow metrics and validated group differences with pairwise t-tests to support Latent Class Analysis.
- **Chinese Economy Sentiment Project** (Supervisor: Dr. Derry Wijaya): Engineered a bilingual (Chinese/English) media-analytics pipeline over 1M+ news articles; fine-tuned BERT on HPC clusters for relevance and sentiment classification, achieving 90% accuracy with 5-fold cross-validation.

Other Internships

- Algorithm Engineer Intern at Cloudwalk, Inc. (Jun. 2019 – Aug. 2019, Shanghai, China): facial recognition model optimization.
- Data Engineer Intern at IBM China (Jun. 2018 – Aug. 2018, Beijing, China): chatbot sentiment analysis.

Technical Skills

Languages: Python, SQL, R, LaTeX
Frameworks & Tools: PyTorch, HuggingFace Transformers, spaCy, LangChain, MongoDB, Git, Docker, Linux/HPC, Prodigy, Doccano